

1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

The image shows a Wireshark network traffic capture. The packet list pane displays a list of captured packets. The packet details pane shows the structure of the selected packet (Frame 817), which is a DNS query. The packet bytes pane shows the raw data of the selected packet.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
812	15.426432500	40.126.17.132	192.168.1.4	TCP	58	443 → 20267 [ACK] Seq=4008 Ack=7939 Win=4194816 Len=0
813	15.431134300	2401:4900:8f73:90c5::2603:1063:5:113::36	2401:4900:8f73:90c5::2603:1063:5:113::36	TCP	74	20266 → 443 [FIN, ACK] Seq=3171 Ack=9350 Win=64768 Len=0
814	15.478858200	2603:1063:5:113::36	2401:4900:8f73:90c5::2603:1063:5:113::36	TCP	78	443 → 20266 [ACK] Seq=9350 Ack=3172 Win=65280 Len=0
815	15.478859000	2603:1063:5:113::36	2401:4900:8f73:90c5::2603:1063:5:113::36	TCP	78	443 → 20266 [FIN, ACK] Seq=9350 Ack=3172 Win=65280 Len=0
816	15.478901100	2401:4900:8f73:90c5::2603:1063:5:113::36	2401:4900:8f73:90c5::2603:1063:5:113::36	TCP	74	20266 → 443 [ACK] Seq=3172 Ack=9351 Win=64768 Len=0
817	15.694965600	fe80::ac07:9e95:6e2::fe80::4663:c2ff:fe34::3	fe80::4663:c2ff:fe34::3	DNS	96	Standard query 0x99a0 HTTPS www.flipkart.com
818	15.695380500	fe80::ac07:9e95:6e2::fe80::4663:c2ff:fe34::3	fe80::4663:c2ff:fe34::3	DNS	96	Standard query 0x7aa9 AAAA www.flipkart.com
819	15.695712100	fe80::ac07:9e95:6e2::fe80::4663:c2ff:fe34::3	fe80::4663:c2ff:fe34::3	DNS	96	Standard query 0x9e2e A www.flipkart.com
820	15.704288000	40.126.17.132	192.168.1.4	TCP	1308	443 → 20267 [ACK] Seq=4008 Ack=7939 Win=4194816 Len=1250 [TCP PDU reassembled in 834]
821	15.704289300	40.126.17.132	192.168.1.4	TCP	1308	443 → 20267 [ACK] Seq=5258 Ack=7939 Win=4194816 Len=1250 [TCP PDU reassembled in 834]
822	15.704290000	40.126.17.132	192.168.1.4	TCP	1308	443 → 20267 [ACK] Seq=6508 Ack=7939 Win=4194816 Len=1250 [TCP PDU reassembled in 834]
823	15.704290900	40.126.17.132	192.168.1.4	TCP	1308	443 → 20267 [ACK] Seq=7758 Ack=7939 Win=4194816 Len=1250 [TCP PDU reassembled in 834]
824	15.704291800	40.126.17.132	192.168.1.4	TCP	1308	443 → 20267 [ACK] Seq=9008 Ack=7939 Win=4194816 Len=1250 [TCP PDU reassembled in 834]
825	15.704292500	40.126.17.132	192.168.1.4	TCP	1308	443 → 20267 [ACK] Seq=10258 Ack=7939 Win=4194816 Len=1250 [TCP PDU reassembled in 834]
826	15.704293100	40.126.17.132	192.168.1.4	TCP	1308	443 → 20267 [ACK] Seq=11508 Ack=7939 Win=4194816 Len=1250 [TCP PDU reassembled in 834]
827	15.704293800	40.126.17.132	192.168.1.4	TCP	1308	443 → 20267 [ACK] Seq=12758 Ack=7939 Win=4194816 Len=1250 [TCP PDU reassembled in 834]

Packet Details:

- Frame 817: Packet, 96 bytes on wire (768 bits), 96 bytes captured (768 bits) on interface \Device\NPF...
- Ethernet II, Src: Intel_ca:88:f2 (30:f6:ef:ca:88:f2), Dst: ZyxelCommuni_34:3c:c0 (44:63:c2:34:3c:c0)
- Internet Protocol Version 6, Src: fe80::ac07:9e95:6e2e:6aca, Dst: fe80::4663:c2ff:fe34:3cc0
- User Datagram Protocol, Src Port: 65419, Dst Port: 53
- Domain Name System (query)
 - Transaction ID: 0x99a0
 - Flags: 0x0100 Standard query
 - Questions: 1
 - Answer RRs: 0
 - Authority RRs: 0
 - Additional RRs: 0
 - Queries
 - www.flipkart.com: type HTTPS, class IN
 - Name: www.flipkart.com
 - Name Length: 16
 - Label Count: 3
 - Type: HTTPS (65) (HTTPS Specific Service Endpoints)

Packet Bytes:

0000 44 63 c2 34 3c c0 30 f6 af ca 88 f2 86 dd 60 08 Dc 4c 00
0010 65 05 00 2a 11 40 fe 80 00 00 00 00 00 00 ac 07 e- 8-
0020 9e 95 6e 2e 6a ca fe 80 00 00 00 00 00 00 46 63 n.jFC
0030 c2 ff fe 34 3c c0 ff 8b 00 35 00 2a 65 2b 99 a05.g+++
0040 01 00 00 01 00 00 00 00 00 00 03 77 77 77 08 667777.8.66
0050 6c 69 70 66 61 72 74 03 63 6f 6d 00 00 41 00 01 lipkart-.com-A..

2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

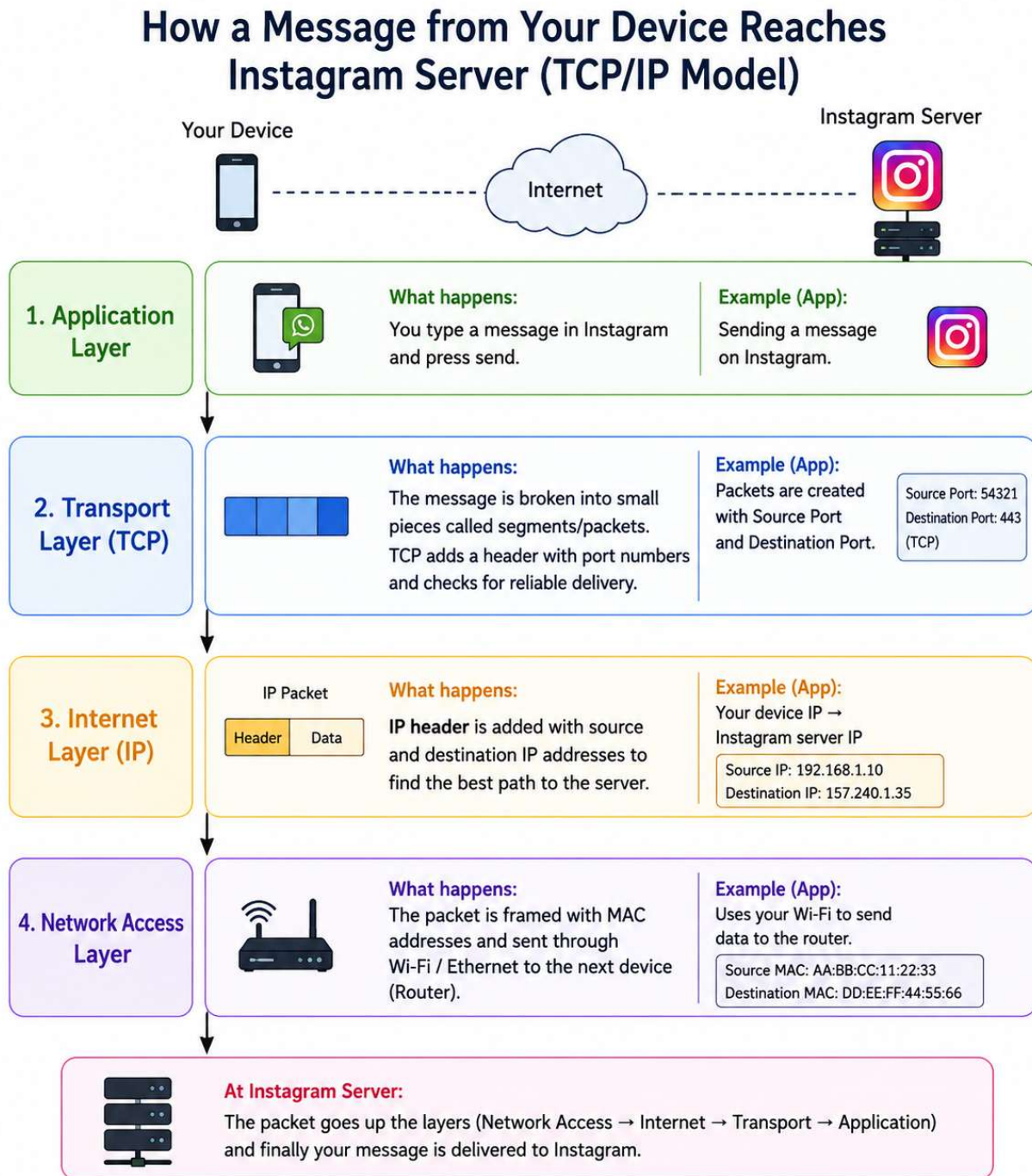
OSI Layer	TCP/IP Layer	Real-Life Example
Application	Application	Sending a WhatsApp message or placing a Zomato order
Presentation	Application	Message encryption and formatting
Session	Application	Keeping the WhatsApp or Instagram session active
Transport	Transport	Splits data into packets using TCP
Network	Internet	Adds IP address to send data to the correct device
Data Link	Network Access	Uses MAC address to send data within the local network
Physical	Network Access	Transfers data through Wi-Fi or Ethernet cable

3. Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.

Field	Work
Source Port	Shows the sender's port number.
Destination Port	Shows the receiver's port number.
Sequence Number	Keeps packets in the correct order.
Acknowledgement Number	Confirms that data has been received.
Header Length	Shows the size of the TCP header.
Flags	Controls the TCP connection (SYN, ACK, FIN, etc.).
Window Size	Shows how much data can be received at one time.
Checksum	Checks whether the data has errors.
Urgent Pointer	Indicates urgent data (rarely used).

4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).



5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

- **Packet**

Protocol: TCP

- **TCP/IP Layer**

Transport Layer

- **How I Identified It**

The Protocol column in Wireshark showed TCP, and the packet contained Source Port and Destination Port fields.

- **Real-World Example**

This packet represents a user sending a WhatsApp message. TCP ensures that the message packets are delivered correctly and in the correct order.

4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

How a message from my device reaches Instagram's server (Through TCP/IP Layers)

